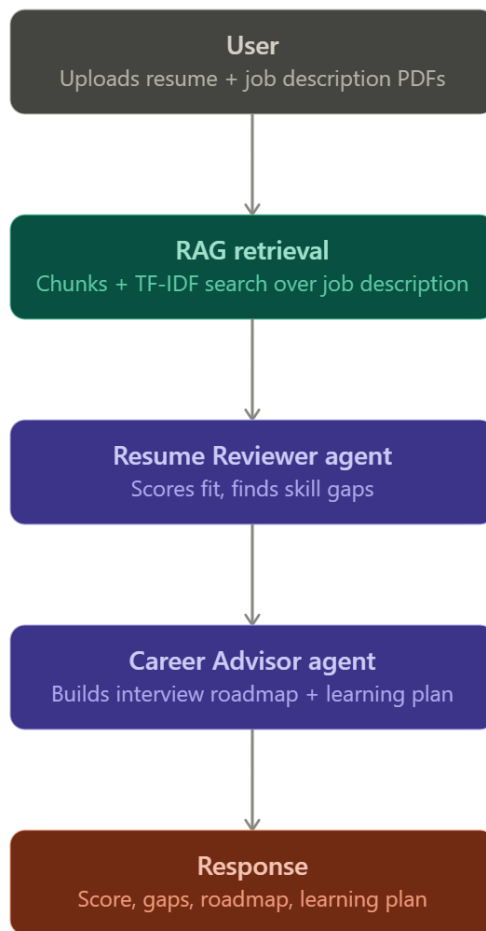


The Architecture Diagram.



What is included

ResumeAI_Project/

|

```

├── app.py                # FastAPI backend + RAG + analysis + agent
logic
├── requirements.txt      # Python dependencies
├── .env.example          # Optional LLM configuration
├── .gitignore
├── README.md             # Complete GitHub README
├── PROJECT_BRIEF.txt     # Short project description
├──
├── uploads/
├──
├── static/
│   ├── index.html        # Website UI
│   ├── style.css         # Responsive styling
│   └── app.js            # Frontend functionality

```

Project Overview

ResumeAI — AI Resume Reviewer & Career Advisor

ResumeAI is a web-based AI career assistant that helps students tailor their resumes for specific job roles.

The user can:

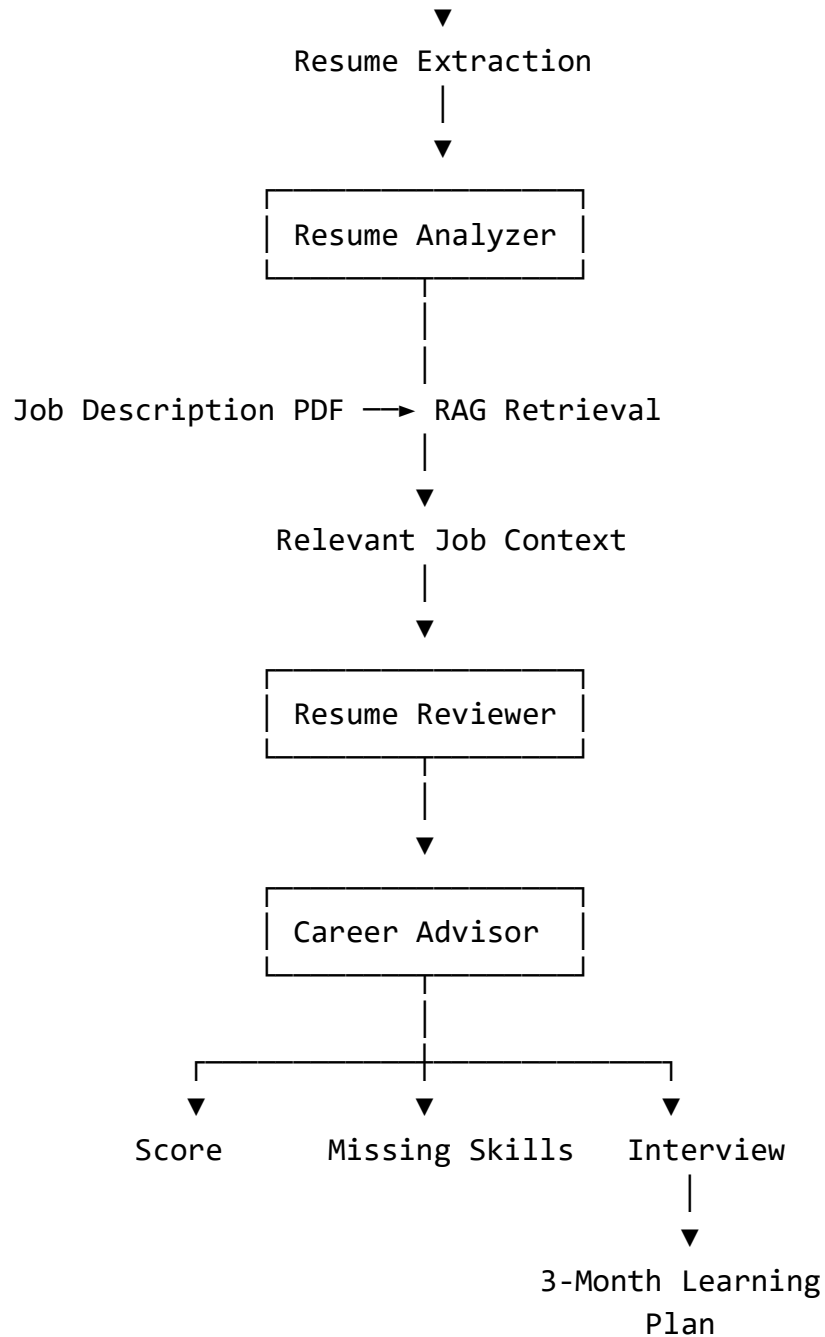
1. Upload a **resume PDF**
2. Paste resume text
3. Enter a target job role
4. Upload **job-description PDFs**
5. Use RAG to retrieve relevant job requirements
6. Analyze resume-job alignment
7. Identify missing skills
8. Receive resume-review feedback
9. Get an interview preparation roadmap
10. Generate a personalized **3-month learning plan**

Core Architecture

```

Resume PDF
|

```



RAG Implementation

The prototype implements:

PDF → Text → Chunking → TF-IDF → Cosine Similarity → Relevant Context

This keeps the project lightweight and lets you run it locally without needing a vector database or API key.

Agent Components

Resume Reviewer

- Checks role alignment
- Finds matching skills
- Identifies skill gaps
- Provides resume improvement suggestions

Career Advisor

- Prioritizes missing skills
- Creates interview preparation steps
- Creates the 3-month learning roadmap

Website Output

The dashboard displays:

- **Resume Score /100**
- Alignment verdict
- Matched skills
- Missing skills
- Resume reviewer feedback
- Career advisor recommendations
- Interview preparation roadmap
- RAG-retrieved job-description context
- 3-month personalized learning plan